



Web: inceptez.com Mail: info@inceptez.com Call: 7871299810, 7871299817

Power BI



Transform in Power BI

Transform in Power BI means **cleaning, shaping, and preparing your raw data** before creating reports. This is done using the **Power Query Editor**.

1. Data Cleaning

- Remove null/blank rows
- Remove duplicates
- Rename columns

2. Change Data Types

Convert columns to:

- Text
- Number
- Date

Ensures accurate calculations

3. Filter & Sort

- Filter rows (e.g., only 2025 data)
- Sort values (ascending/descending)

4. Split & Merge Columns

- Split (e.g., Full Name → First/Last)
- Merge multiple columns into one

5. Add New Columns

- Conditional column (using formulas)
- Logical column (like IF logic)

6. Group By (Aggregation)

Summarize data:

- Sum
- Count
- Average

7. Pivot / Unpivot

- **Pivot** → rows to columns
- **Unpivot** → columns to rows

Chart Types in Power BI

In **Microsoft Power BI**, charts (visuals) are used to represent data in a meaningful and interactive way.

Most important chart types:

1. Bar & Column Charts

2. Line Charts

3. Pie & Donut Charts

4. Maps

5. Table & Matrix

6. KPI & Card Visuals

7. Scatter & Bubble Charts

8. Area Charts

9. Funnel Charts

10. Waterfall Charts

Hierarchies

1. Power BI Desktop

A **Windows application** used for building reports and data models

- Connect to data sources (Excel, SQL, web, etc.)
- Transform data using **Power Query**
- Create relationships and calculations (DAX)
- Design interactive reports and visuals

This is where most development work happens

2. Power BI Service: A cloud-based platform

- Publish reports created in Desktop
- Share dashboards with others
- Collaborate via workspaces
- Schedule data refresh

Ideal for **team collaboration and sharing insights**.

3. Power BI Mobile Apps: Available on iOS, Android, tablets

View dashboards on the go

Receive alerts and notifications

Interact with reports

Great for **executives and field users**.

Power BI Service

Power BI Service is the cloud-based part of Power BI that **publish, share, and collaborate on reports and dashboards**

Key Features

Dashboards

- Create interactive dashboards with multiple visuals
- Combine data from different reports into one view
- Real-time updates using live data

Reports Sharing & Collaboration

- Share reports with individuals or teams
- Control access (view/edit permissions)
- Collaborate using workspaces

Workspaces

- Organize reports, dashboards, and datasets
- Enable team-based development
- Manage roles: Admin, Member, Contributor, Viewer

Data Refresh (Scheduled & Automatic)

- Schedule refresh (daily, hourly)
- Connect to on-prem data using gateway
- Keep reports always up-to-date

Dataflows & Datasets

- Centralized data preparation (ETL-like)
- Reuse datasets across multiple reports
- Supports scalable data modeling

Real-Time Streaming

- Stream live data (IoT, logs, events)
- Instant dashboard updates
- Useful for monitoring systems

Power BI: PBIX vs PBIT

Feature	PBIX	PBIT	PBIP
Full Form	Power BI Report File	Power BI Template File	Power BI Project File
Purpose	Complete report with data	Reusable template without data	Developer-friendly project structure
Contains Data	Yes	No	No (references datasets/files)
Contains Report Design	Yes	Yes	Yes
Contains Queries & Model	Yes	Yes	Yes
File Type	Single binary file	Template file	Folder-based project
Used For	Report development & sharing	Creating reusable templates	Source control & team collaboration
Suitable for Git	Difficult	Difficult	Best suited
Editable in Text Format	No	No	Yes (many files are text-based)
Introduced For	Standard reporting	Reusable templates	Modern Power BI DevOps
Extension	.pbix	.pbit	.pbip
Typical Use Case	Build and publish reports	Share report structure without sensitive data	Collaborative development with version control

Power BI: Calculated vs Measure

Feature	Calculated Column	Measure
Calculation Level	Row-level	Aggregate level
Storage	Stored in model	Not stored
Performance	Uses more memory	Efficient
Context	Row context	Filter context
Usage	Columns, slicers, filters	Values in visuals
Recalculation	During data refresh	Dynamic (real-time)

Power BI: Gateways

A **Power BI Gateway** is a bridge that securely connects on-premises data sources with Microsoft cloud services such as:

- Power BI Service
- Power Apps
- Power Automate
- Azure Analysis Services

It allows Power BI Service to access data that is stored inside a private network or data center.

Why Gateway is Needed?

Power BI Service runs in the cloud.

If data is stored:

- On-premises SQL Server
- Oracle Database
- Local Excel files
- SAP systems
- MySQL server
- File shares

then Power BI cloud service cannot directly access them securely.

The gateway acts as a secure communication channel.

Type	Description	Usage
Personal Mode Gateway	Installed for single user	Personal refresh
Standard / Enterprise Gateway	Shared gateway for multiple users	Enterprise reporting

Power BI: Reports vs Dashboard

Feature	Power BI Report	Power BI Dashboard
Definition	A collection of multiple visualizations/pages created from datasets	A single-page summary view containing important visuals
Pages	Multiple pages supported	Only one page
Data Source	Can use one or multiple datasets	Uses visuals pinned from reports
Interactivity	Highly interactive with filters, slicers, drill-through, bookmarks	Limited interaction
Creation Tool	Mainly created in <u>Power BI Desktop</u> and published to <u>Power BI Service</u>	Created only in <u>Power BI Service</u>
Purpose	Detailed analysis and exploration	High-level monitoring and quick insights
Visualization Layout	Structured report pages	Tile-based layout
Drill Down Capability	Supported	Limited
Export Options	PDF, PPT, Excel, etc.	Limited export
Real-time Monitoring	Possible but less common	Commonly used for real-time KPI monitoring
Use Case	Business analysts exploring data deeply	Managers/executives tracking KPIs